

SAMARTH CHAUHAN

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EXPERIENCE

Backend and AI Intern

Jun 2025 – Present

Motorola Solutions

- Built the cloud-based **Agent Playground** (shipped inside internal **AI Foundry**), letting developers and PMs connect live to **MCP Servers**, **A2A-compatible Agents** and configs in real time to inspect agent behavior.
- Implemented isolated **multi-user session environments** ($N \text{ users} \times M \text{ MCP servers}$) with per-session runtime using **Python**, **Google ADK**, and GCP Vertex AI SessionService — deployed across internal engineering and product teams.
- Contributed to an enterprise **RAG**-based Google Chat agent (Node.js, **AWS Bedrock**, AWS S3); owned the real-time interaction layer, streaming live agent reasoning states (thinking, searching) to the client and rendering parsed markdown responses.
- Developed an authorization middleware microservice in **TypeScript** securing internal **AI Foundry** infrastructure, handling complex **OAuth 2.0** and **Gated DCR OAuth** flows with **Redis** session caching and **MongoDB** persistent state.
- Evaluated frameworks **Google ADK**, **CrewAI**, **AutoGen** and architectures incorporating **MCP** and **Agent Harness**; findings directly informed the stack selection (ADK + Vertex AI SessionService) for the Agent Playground.

Flutter Developer Intern

Nov 2024 – May 2025

Employee Forums

- Architected and migrated a legacy Dart codebase to the **BLoC** pattern using comprehensive dependency injection.
- Optimized cold startup time from **7s to 2s** (**71% improvement**) by implementing lazy API invocation and deferred module loading.
- Designed robust request/response data models and implemented **SWR** caching for seamless REST API integration.

Flutter Developer Intern

May 2024 – Aug 2024

Xingoda

- Integrated Firebase backend services and **RazorPay** payment gateway into a Dart/Flutter application for secure transaction processing.
- Optimized state management and API integration workflows across core application modules.

PROJECTS

VECSS - Distributed Video Transcoding System | Golang, Python, Docker, RabbitMQ, FFmpeg

- Designed and implemented a scalable microservices-based platform for video transcoding using **Golang** and **Python**.
- Built an auto-scaling pipeline utilizing **RabbitMQ** and S3-compatible storage for asynchronous video processing.
- Containerized workloads with **Docker** and orchestrated **FFmpeg** workers to concurrently transcode videos into multiple resolutions.

Resume Cartel - Multi-Agent Resume Builder | Python, Google ADK, Redis, Gemini

- Built a hierarchical multi-agent pipeline in **Python** using **Google ADK** — a root Coach agent that delegates to Writer, Critic, and YAML sub-agents, each reading and writing shared versioned markdown artifacts as inter-agent state.
- Backed agent memory with **Redis**; Agents write structured facts, notes, and file artifact references into a shared key-value store, giving sub-agents (Writer, Critic) read access to accumulated context without passing it through the LLM prompt each turn.
- Structured final output as typed YAML consumed by a downstream template engine, separating the LLM reasoning layer from document rendering and enabling the same data to drive multiple output formats.

Xginx - Layer 7 Load Balancer | Golang

- Developed a custom **Layer 7** HTTP load balancer from scratch in **Golang**, engineering robust session-based routing and load distribution algorithms.
- Implemented automated, continuous health checks and retry mechanisms to actively monitor backend server pools, ensuring high availability and robust fault tolerance.
- Architected custom reverse proxy request handling to efficiently manage concurrent client connections.

SKILLS

Languages: Golang | Python | TypeScript(Node.js) | Dart

AI & Agentic: Google ADK | Model Context Protocol (MCP) | Agent-to-Agent (A2A) | AWS AgentCore | GCP Vertex AI

Backend & Infra: Docker | AWS S3 | AWS Lambda | GCP Cloud Run | Redis | MySQL | RabbitMQ | MongoDB

Frameworks: Gorilla/Mux | FastAPI | Flutter

EDUCATION

Charotar University of Science and Technology (CHARUSAT)

B.Tech in Information Technology, CGPA 9.14

Anand, Gujarat

Sep 2022 – Present

Shri Ramkrishna Harikrishna Academy

Higher Secondary Certificate (CBSE), 90%

Surat, Gujarat

Jan 2020 – May 2022

ACHIEVEMENTS

- **Winner**, State Level Project Competition at Vidush Somany Institute.
- **Winner**, SCIENCE Manthan Competition at P. D. Patel Institute.

RESPONSIBILITIES

Tech Lead **Mar 2024 – Dec 2024**

CP Squad Club, CHARUSAT

- Orchestrated 5 technical contests with average participation of 400-500 students per contest, while managing competitive programming platforms.